7.7 Practice: metric system and area

← Unit 7.7 · Answers

The ladder, from big to small: **km – hm – dam – m – dm – cm – mm** *Kan Het Dametje Met De Centimeter Meten?* Each step down: × 10. Each step up: : 10.

Draw the shapes on squared paper whenever it helps.

Set 1 · Warm-up: lengths

1. 1 m = … cm
2. 1 km = … m
3. 1 cm = … mm
4. 3 m = … cm
5. 2 km = … m
6. 50 mm = … cm
7. 400 cm = … m
8. 1 m = … dm
9. Half a metre = … cm

Talk about it: In question 7, did the number get bigger or smaller? Why?

Set 2 · Building: converting with kommagetallen

1. 1,5 m = … cm
2. 2,75 m = … cm
3. 3 500 m = … km
4. 85 cm = … m
5. 1,2 km = … m
6. 25 mm = … cm
7. 4 dm = … cm
8. 250 ml = … l
9. 1,5 kg = … g
10. 750 g = … kg
11. 3 ons = … g
12. 1 l = … ml

Talk about it: For question 3, how many steps on the ladder are there from m to km?

Set 3 · Building: perimeter (*omtrek*)

1. A rectangle is 5 cm long and 3 cm wide. What is its omtrek?
2. A square has sides of 4 m. What is its omtrek?
3. A garden is 10 m long and 6 m wide. What is its omtrek?
4. A rug is 2,5 m long and 1,5 m wide. What is its omtrek?
5. A square has an omtrek of 20 cm. How long is each side?
6. A rectangle has an omtrek of 30 m. It is 10 m long. How wide is it?
7. A garden is 12 m long and 8 m wide. You put a fence all the way round.
   a. How many metres of fence do you need?
   b. Fencing costs € 15 per metre. What does the fence cost?
8. A square has sides of 1 m. What is its omtrek?

Talk about it: How could you explain “omtrek” to a younger child?

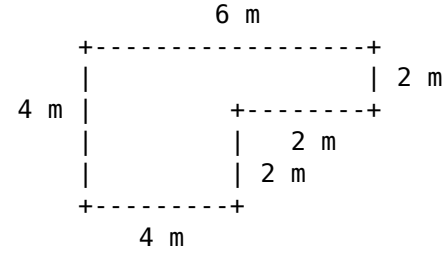
Set 4 · Building: area (*oppervlakte*)

1. A rectangle is 3 cm long and 2 cm wide. What is its oppervlakte?
2. A square has sides of 5 m. What is its oppervlakte?
3. A vegetable patch is 8 m long and 4 m wide. What is its oppervlakte?
4. A tile is 12 cm long and 10 cm wide. What is its oppervlakte?
5. A bedroom is 4 m long and 3,5 m wide. What is its oppervlakte?
6. A rectangle has an oppervlakte of 24 cm². It is 6 cm long. How wide is it?
7. A square has an oppervlakte of 49 m². How long is each side?
8. A living-room floor is 5 m by 4 m. Laminate flooring costs € 12 per m². What does the floor cost?
9. Rectangle A is 6 m by 6 m. Rectangle B is 9 m by 4 m.
   a. Which has the bigger oppervlakte?
   b. Which has the bigger omtrek?
10. A square has sides of 10 m. What is its oppervlakte?

Talk about it: Question 9 is surprising. What does it tell you about omtrek and oppervlakte?

Set 5 · Stretch: composite shapes and m²/cm²

Shape P is an L-shaped room. It’s a 6 m × 4 m rectangle with a 2 m × 2 m square cut out of one corner.



1. What is the oppervlakte of shape P?
2. What is the omtrek of shape P? (Walk round all six sides.)

Shape Q is two rectangles side by side: one is 5 m × 3 m, the other 4 m × 2 m.

3. What is the total oppervlakte of shape Q?

More:

4. A garden is 10 m by 8 m. There is a shed of 3 m by 2 m in it. How many m² of grass are there?
5. A photo frame is 30 cm by 20 cm on the outside. The photo inside is 26 cm by 16 cm. What is the oppervlakte of the frame itself (the border)?
6. 1 m² = … cm² (Hint: how many cm along each side of a 1 m square?)
7. How many tiles of 10 cm × 10 cm fit on 1 m²?
8. A floor is 3 m by 2 m. You use square tiles of 50 cm × 50 cm. How many tiles do you need?
9. A rectangle is 2 m by 3 m. What is its oppervlakte?

Talk about it: In question 2, cutting the corner out made the area smaller. Did the omtrek change? Why?

Puzzle corner (optional)

The best rabbit run. You have 12 metres of fence to make a rectangular rabbit run. The sides have to be whole metres.

- What different rectangles can you make? (Draw them on squared paper.)
- Which one gives the rabbit the most space (oppervlakte)?
- What if you had 20 metres of fence? Can you predict the best shape before trying?
- What if one side is against the house wall and doesn’t need a fence?